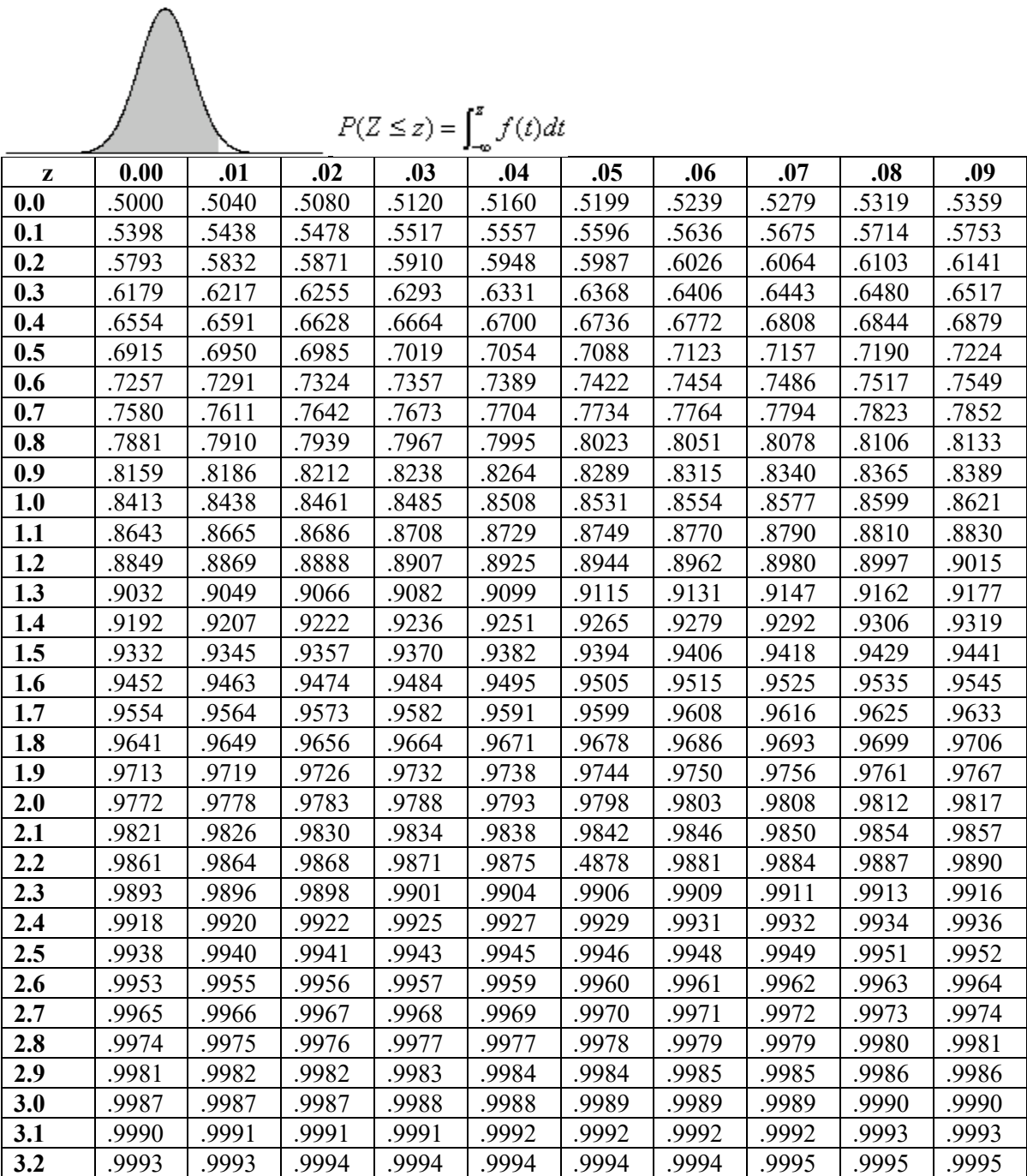


Tabla de distribución normal estándar. La tabla de normalidad es simétrica. Por ende $P(Z > -x) = P(Z < x)$.



Formulario

$$CT = DC + \frac{D}{Q}S + \frac{Q}{2}H$$
$$Q_{eq} = \sqrt{\frac{2 \times D \times S}{H}}$$
$$R = d \times L$$
$$R = d \times L + z_{\alpha} \sigma \sqrt{L}$$

$$Q^* = F^{-1}\left(\frac{c_u}{c_o + c_u}\right)$$
$$\mu = \frac{a + 4m + b}{6} \qquad \sigma = \frac{b - a}{6}$$

$$EF = ES + t$$
$$LS = LF - t$$
$$Z = \frac{D - T_E}{\sqrt{\sum \sigma_i^2}}$$

$$Q^* = d \times (T + L) + z_{\alpha} \sigma \sqrt{(T + L)} - I_{\text{existente}}$$
$$I = T_p \times (p - d)$$

$$Q = T_p \times p$$
$$T_p = \frac{Q}{p}$$

$$ef = \frac{k}{k + 1}$$

$$P = \sqrt{\left(\frac{a}{2}\right) \left(\frac{q}{z}\right)} \quad P = \sqrt{\left(\frac{a}{2}\right) \left(\frac{1}{n}\right) \left(\sum_{i=1}^n \frac{q_i}{z_i}\right)}$$

$$Ben = sp_i - c_r d_i$$
$$Ben = s(p_i + D_i)$$

$$Beneficio_{min_A} = \frac{s * p_i - c_r * d_i}{l_i}$$
$$Restocks = \frac{f_i}{V_i} \text{ Restocks/tiempo}$$

$$Beneficio_{adic_A} = \frac{s * D_i + c_r * d_i}{u_i - l_i}$$
$$\text{Costo Total} = \text{Costo Fijo} + \text{Costo Variable} \times \text{Volumen}$$

$$v_i^* = \left(\frac{\sqrt{f_i}}{\sum_{j=1}^n \sqrt{f_j}} \right) \vee \frac{p_i}{\sqrt{f_i}}$$
$$C_x = \frac{\sum d_{ix} V_i}{\sum V_i} \quad C_y = \frac{\sum d_{iy} V_i}{\sum V_i}$$

$$\rho = \frac{\lambda}{\mu} \quad \rho = \frac{\lambda}{c\mu} \quad L = \lambda \times W \quad c_T = \frac{\sigma}{t} = \frac{\sqrt{\text{Var}(T)}}{E(T)}$$

$$L = \frac{\rho}{1 - \rho}, \quad W = \frac{1}{\mu(1 - \rho)} \quad L_q = \frac{\rho^2}{1 - \rho}, \quad W_q = \frac{\rho}{\mu(1 - \rho)}$$

$$WIP = TH \times TC \quad L = \lambda * W$$

$$A = \frac{m_f}{m_r + m_f} \quad t_e = \frac{t_o}{A} \quad \sigma_e^2 = \left(\frac{\sigma_o^2}{A}\right) + \frac{(m_r + \sigma_r^2)(1 - A)t_o}{Am_r}$$

$$c_e^2 = \frac{\sigma_e^2}{t_e^2} = c_o^2 + (1 + c_r^2)A(1 - A)\frac{m_r}{t_o}$$

$$t_e = t_o + \frac{t_s}{N_s} \quad \sigma_e^2 = \sigma_o^2 + \frac{\sigma_s^2}{N_s} + \frac{N_s - 1}{N_s^2} t_s^2 \quad c_e^2 = \frac{\sigma_e^2}{t_e^2}$$

$$(c_s)^2 \approx \rho^2 (c_e)^2 + (1 - \rho^2) (c_a)^2 \quad CT_q = \underbrace{\left(\frac{c_a^2 + c_e^2}{2}\right)}_V \underbrace{\left(\frac{\rho}{1 - \rho}\right)}_U \underbrace{t_s}_T$$

$$L = \frac{\rho}{1 - \rho} - \frac{(b + 1)\rho^{b+1}}{1 - \rho^{b+1}}$$

$$\lambda' = \lambda \left(\frac{1 - \rho^b}{1 - \rho^{b+1}} \right)$$

$$L_q = \frac{\rho}{1 - \rho} \times Prob(N > c)$$

$$W_q = \frac{\rho}{\lambda(1 - \rho)} \times Prob(N > c)$$

$$F_t = w_1A_{t-1} + w_2A_{t-2} + w_3A_{t-3} + \dots + w_nA_{t-n}$$

$$T_{(t,t-1)} = A_t - A_{t-1}$$

$$F_{t+1} = \alpha A_t + (1 - \alpha)F_t$$

$$\hat{y} = a + bx$$

$$\sum_1^n w_n = 1$$

$$T = \frac{\sum_{i=1}^n T_{(t-i,t-i-1)}}{n}$$

$$TS_k = \frac{\sum_{i=1}^k e_k}{MAD_k}$$

$$F_t = \frac{A_{t-1} + A_{t-2} + A_{t-3} + \dots + A_{t-n}}{n}$$

$$e_t = F_t - A_t$$

$$\sigma = 1,25 * MAD$$

$$MAD_k = \frac{1}{k} \sum_{i=1}^k |e_i|$$

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$
$$a = \frac{\sum y}{n} - b \frac{\sum x}{n} = \bar{y} - b\bar{x}$$

$$FIT_t = \underbrace{F_t}_{\text{Pronóstico}} + \underbrace{T_t}_{\text{Tendencia}}$$

$$F_t = FIT_{t-1} + \alpha(A_{t-1} - FIT_{t-1})$$

$$T_t = T_{t-1} + \alpha\delta(A_{t-1} - FIT_{t-1})$$

$$L = t_k + t_p + t_v$$

$$LCS = \bar{x} + Z * \sigma$$
$$LCI = \bar{x} - Z * \sigma$$

$$N = \frac{D \times L}{C}$$

$$N = \frac{D \times L}{C} (1 + \varepsilon)$$

Planes de muestreo para alpha = 0.05 y beta = 0.10

c	LTPD/AQL	n*AQL
0	44,890	0,052
1	10,946	0,355
2	6,509	0,818
3	4,890	1,366
4	4,057	1,97
5	3,549	2,613
6	3,206	3,286
7	2,957	3,981
8	2,768	4,695
9	2,618	5,426

$$p_0 \pm 3 \sqrt{\frac{p_0(1-p_0)}{n}}$$

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$LCS \bar{X} = \bar{\bar{X}} + A_2 * \bar{R}$$

$$LCI \bar{X} = \bar{\bar{X}} - A_2 * \bar{R}$$

$$LCS R = D_4 * \bar{R}$$

$$LCI R = D_3 * \bar{R}$$

$$C_{pk} = \frac{USL - Media}{3\sigma}$$

$$C_{pk} = \frac{Media - LSL}{3\sigma}$$

Tamano Muestra	A ₂	d ₂	D ₃	D ₄
2	1.880	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.282
5	0.577	2.326	0	2.114
6	0.483	2.534	0	2.004
7	0.419	2.704	0.076	1.924
8	0.373	2.847	0.136	1.864
9	0.337	2.970	0.184	1.816
10	0.308	3.078	0.223	1.777
11	0.285	3.173	0.256	1.744
12	0.266	3.258	0.283	1.717
13	0.249	3.336	0.307	1.693
14	0.235	3.407	0.328	1.672
15	0.223	3.472	0.347	1.653
16	0.212	3.532	0.363	1.637
17	0.203	3.588	0.378	1.622
18	0.194	3.640	0.391	1.608
19	0.187	3.689	0.403	1.597
20	0.180	3.735	0.415	1.585
21	0.173	3.778	0.425	1.575
22	0.167	3.819	0.434	1.566
23	0.162	3.858	0.443	1.557
24	0.157	3.895	0.451	1.548
25	0.153	3.931	0.459	1.541